# Global research status and hotspots in telecare applications during 2003-2022: A bibliometric and visualized analysis

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## ABSTRACT

Objective: This study aims to explore the status of research and hot topics in the field of telecare and provide a theoretical basis for future research development in this field. Methods: The literature on telecare in the Web of Science core collection was searched and analyzed using CiteSpace software for visualization and analysis. Results: A total of 1267 papers were included in the study. The number of annual publications in telecare research has consistently increased over the past 20 years. The United States has the highest number of publications in this field. Khan, Muhammad Kashif Iqbal is the author with the highest number of publications. Research institutions primarily consist of universities. The relevant research findings are predominantly published in journals specializing in healthcare science and services. The main research topics in telecare include telecare technology optimization, telecare application areas, and telecare goal evaluation. Conclusion: Existing research strives to enhance user information security and improve the efficiency and quality of healthcare by exploring innovative user authentication schemes and federated artificial intelligence technologies. The COVID-19 pandemic has accelerated the advancement of telecare, with emergency medicine and chronic disease management being its primary application areas. Enhancing patient self-care stands as a key objective within this domain.

## CCS CONCEPTS

• **Social and professional topics**; • **Computing / technology policy**; • **Medical information policy**; • **Medical technologies**; • **Remote medicine**; • **Networks**; • **Network protocols**; • **Information systems**; • **Information systems applications**;

## KEYWORDS

Telecare, Telemedicine, Research hotspots, CiteSpace

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## 1 INTRODUCTION

In recent years, the rapid development of technologies such as 5G communication and IoT has led to the emergence of telecare as an innovative solution to effectively alleviate the uneven distribution of healthcare resources, given the context of accelerating global population aging and the increasing burden of chronic diseases [1]. Telecare is an emerging medical model that provides online medical care services to patients at home or in remote areas by large medical

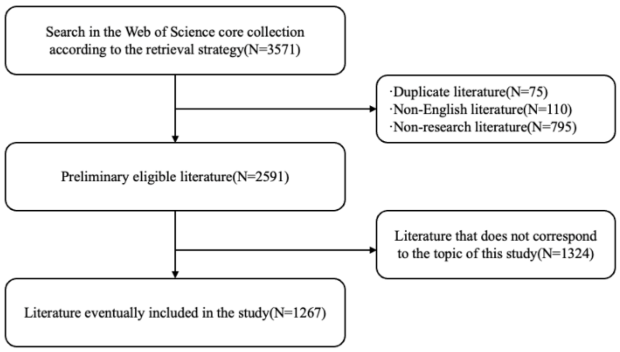


**Figure 1: Flowchart of literature screening**

centers through remote communication, telemetry, and telemetry technologies [2]. Currently, several countries, including the United States, have launched policies and programs to support the application and promotion of telecare. This advancement is expected to significantly reduce medical costs and time, while improving the autonomy and convenience of patients' access to care [3]. Meanwhile, as the industry advances, scientific research continues to emerge. However, the extensive diversity and complexity of the literature make it challenging for researchers to quickly grasp the current research status. Bibliometric analysis is a research method that uses statistical and mathematical methods to quantify and analyze literature. It can transform a large amount of literature data into charts and graphs, thus visually presenting the knowledge structure and correlations of the research field and helping researchers better understand and grasp the core content of the field [4]. Therefore, this study aims to utilize bibliometric methods to visualize and analyze the field of telecare research, providing references and a theoretical basis for its future development.

# 2 MATERIALS AND METHODS

The bibliographic data used for this study were obtained from the Web of Science core collection. The search strategy was as follows: ((((TS=("telenurs*")) OR TS=("telecare")) OR TS=("telehomecare")) OR TS=(telehealthcare)) OR (TS=("telemedicine") AND TS=(nursing)). The search period covered from 2003-01-01 to 2022-12-31. Review papers, conference abstracts, non-English literature, literature not related to the topic of this study, and duplicate publications were removed. A total of 1267 papers were included in this study. To ensure the quality of the selected literature, the literature screening was independently carried out by two researchers who completed the co-checking process. In case of any disputes, the third researcher made the final decision regarding the selection of literature. The specific process of literature screening is shown in Figure 1.

In this study, Citespace 5.7.R2, developed by Dr. Chaomei Chen and based on the Java language, was utilized [5]. The time span was set from 2003 to 2022, and the time Slicing was set to 1 year. The node type selected was "Keyword," and the selection criterion was g-index k=25. The connection strength was set to "Cosine," and the Pruning method used was "Pathfinder."

# 3 RESULTS

## 3.1 Analysis of annual publication volume and country

In this study, Excel was utilized to plot the annual count of publications and the corresponding trend line, as depicted in Figure 2. Over the past 20 years, the number of annual publications in this research field has demonstrated a consistent upward trend. The trend line equation is $y=17e^{0.1071x}$, yielding an $R^2$ value of 0.9622. The US holds the highest number of publications in this research area (n=480), accounting for 37.9% of the total. The combined publications from the top 5 countries represent 67.4% of the total number of publications.

## 3.2 Analysis of author and institutional

The data for authors and institutions were extracted from the Web of Science. The author with the highest number of publications in this research area was Khan, Muhammad Kashif Iqbal (n=12, 0.9%). The author with the highest average citation frequency (AC) was Newman, Stanton (AC=62.78). Table 1 illustrates the publication status of the top 10 authors, collectively contributing to 7.1% of the total literature. The institution with the highest number of publications in this research area was the US Department of Veterans Affairs (n=55, 4.3%). The combined publication volume from the Top 10 institutions accounted for 21.5% of the total literature.

## 3.3 Analysis of journals

The data for journals were extracted from the Web of Science. Table 2 displays the top 10 journals by publication volume. The most published journal in this research area is the Journal of Telemedicine and Telecare (n=192, 15.2%), with seven out of the top 10 journals falling within the Q1 division and seven journals belonging to the Healthcare Sciences and Services category. The combined publications from the top 10 journals represent 60.2% of the total literature.

## 3.4 Analysis of keywords

In this study, a meticulous co-occurrence analysis of keywords within the telecare research domain was undertaken, and the findings are visually presented in Figure 3(A). A total of 596 distinct keyword nodes were meticulously identified in this domain, establishing 2720 keyword links and yielding a Density measure of 0.0153. Excluding common search terms, noteworthy keywords with elevated frequency included "management" (n=151), "technology" (n=115), and "outcome" (n=81). Furthermore, the Latent Dirichlet Allocation (LLR) algorithm was strategically employed to conduct a cluster analysis of keywords in this study, and the outcomes are vividly illustrated in Figure 3(B). The top five keyword clusters unveiled in this research area were categorized as follows: #0 self-care, #1 emergency medicine, #2 authentication, #3 COVID-19, and #4 nurse practitioner. The Modularity Q value stood at 0.4316 (Q > 0.3), and the Mean Silhouette reached an impressive 0.7438 (S > 0.7), affirming that the clustering results effectively encapsulate the prevailing research hotspots within the expansive field of telecare.

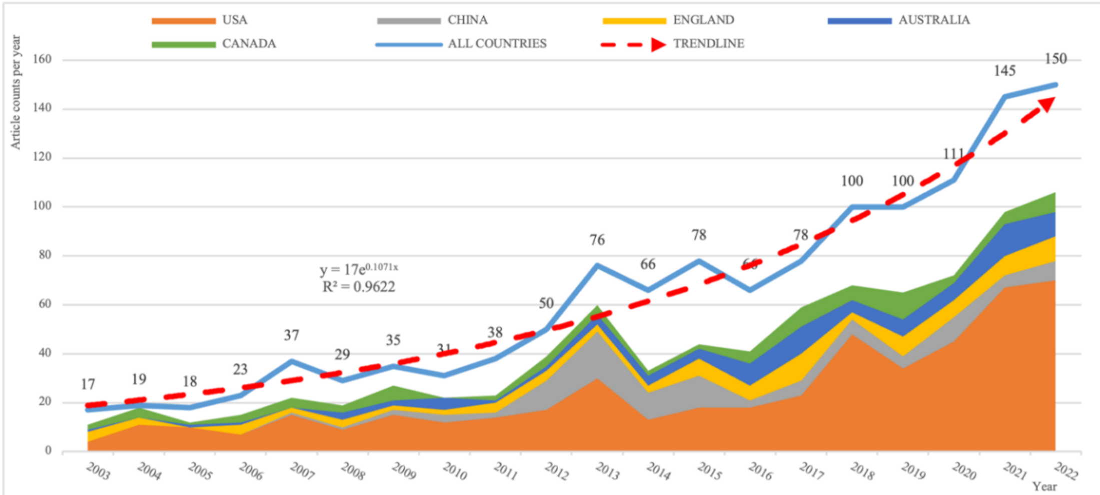


**Figure 2: Mapping of annual publication volume in the field of telecare research over the past 20 years**

**Table 1: Summary of TOP 10 authors and institutions in the field of telecare research**

| Rank | Author | Number of studies (%) | AC[1] | h-index | Rank | Institutions | Number of studies (%) |
|---|---|---|---|---|---|---|---|
| 1 | Khan, Muhammad Kashif Iqbal | 12(0.9%) | 49.25 | 12 | 1 | US DEPARTMENT OF VETERANS AFFAIRS | 55(4.3%) |
| 2 | Scalvini, Simonetta | 10(0.8%) | 22 | 8 | 2 | HARVARD UNIVERSITY | 31(2.4%) |
| 3 | Newman, Stanton | 9(0.7%) | 62.78 | 8 | 3 | UNIVERSITY OF CALIFORNIA SYSTEM | 30(2.4%) |
| 3 | Kroenke, Kurt | 9(0.7%) | 48.78 | 7 | 4 | N8 RESEARCH PARTNERSHIP | 24(1.9%) |
| 3 | Weinstock, Ruth S. | 9(0.7%) | 44.78 | 8 | 4 | UNIVERSITY OF QUEENSLAND | 24(1.9%) |
| 3 | Smith Anthony Carl | 9(0.7%) | 13.78 | 7 | 6 | PENNSYLVANIA COMMONWEALTH SYSTEM OF HIGHER EDUCATION PCSHE | 23(1.8%) |
| 7 | Ole Kristian Hejlesen | 8(0.6%) | 16.25 | 6 | 6 | UNVERSITY OF WASHINGTON | 23(1.8%) |
| 7 | Holmstrom, Inger | 8(0.6%) | 36.63 | 8 | 8 | UPPSALA UNIVERSITY | 22(1.7%) |
| 7 | Bjorkman, Annica | 8(0.6%) | 19.75 | 6 | 9 | UNIVERSITY OF SOUTHERN DENMARK | 21(1.7%) |
| 7 | Frnkelstein, Stanley M. | 8(0.6%) | 40.88 | 7 | 10 | STATE UNIVERSITY SYSTEM OF FLORIDA | 19(1.5%) |

[1]AC: Average citation per article

# 4 DISCUSSION

## 4.1 Analysis of the current state of research

Over the past two decades, there has been a remarkable and consistent increase in the annual publication output within the field of telecare research. This upward trajectory not only underscores the escalating interest of researchers in the advancement and application of telecare technology but also mirrors the surging global demand for telecare solutions. This growing trend is particularly significant as it holds promise in addressing healthcare resource inequalities on a global scale. Among the nations contributing to this surge in publications, the United States emerges as a frontrunner, boasting the highest number of publications in telecare research. This dominance signals the advanced state of development in telecare technology and services within the United States, positioning the country as a benchmark for other nations seeking to embark on similar telecare initiatives. Pioneering scholars such as Khan, Muhammad Kashif Iqbal, and Newman, Stanton have played instrumental roles in shaping the trajectory of telecare research. Khan, Muhammad Kashif Iqbal's noteworthy focus on authentication schemes for users of telemedicine information systems [6] exemplifies a critical aspect of telecare technology, ensuring the secure and reliable access to medical information. On the other hand, Newman, Stanton's impactful studies delve into the intersection of telehealth, diabetes management, and the broader user

**Table 2: Summary of the TOP 10 publications in the field of telecare research**

| Rank | Journal | Number of studies (%) | IF[2](2022) | Category partitioning (2022) |
|---|---|---|---|---|
| 1 | JOURNAL OF TELEMEDICINE AND TELECARE | 192(15.2%) | 4.7 | HEALTH CARE SCIENCES & SERVICES (Q1) |
| 2 | TELEMEDICINE AND E HEALTH | 164(12.9%) | 2.762 | HEALTH CARE SCIENCES & SERVICES (Q1) |
| 3 | JOURNAL OF MEDICAL SYSTEMS | 96(7.6%) | 5.3 | HEALTH CARE SCIENCES & SERVICES (Q1) |
| 4 | TELEMEDICINE JOURNAL AND E HEALTH | 74(5.8%) | 4.7 | HEALTH CARE SCIENCES & SERVICES (Q1) |
| 5 | JOURNAL OF MEDICAL INTERNET RESEARCH | 67(5.3%) | 7.4 | HEALTH CARE SCIENCES & SERVICES (Q1) |
| 6 | BMC HEALTH SERVICES RESEARCH | 58(4.6%) | 2.8 | HEALTH CARE SCIENCES & SERVICES (Q3) |
| 7 | INTERNATIONAL JOURNAL OF MEDICAL INFORMATICS | 35(2.8%) | 4.9 | COMPUTER SCIENCE, INFORMATION SYSTEMS (Q2) |
| 8 | JOURNAL OF CLINICAL NURSING | 34(2.7%) | 4.2 | NURSING (Q1) |
| 9 | HEALTH INFORMATICS JOURNAL | 23(1.8%) | 3 | HEALTH CARE SCIENCES & SERVICES (Q3) |
| 10 | JOURNAL OF ADVANCED NURSING | 20(1.6%) | 3.8 | NURSING (Q1) |

[2]IF: Impact factor

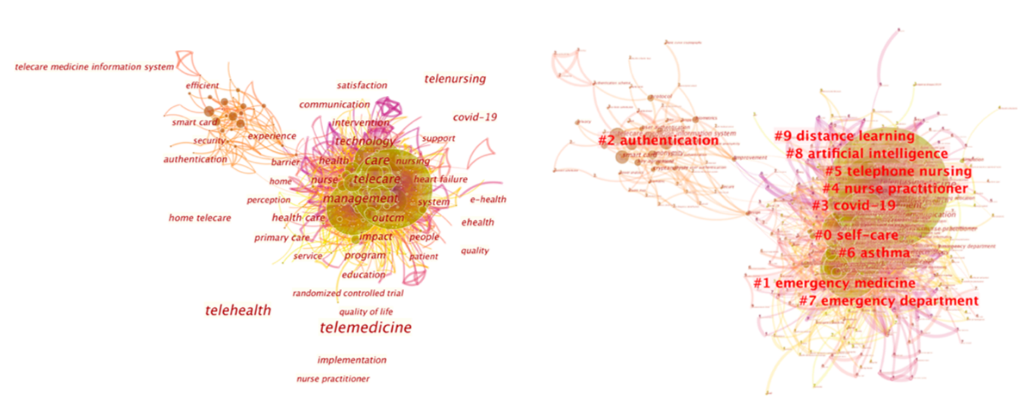


**Figure 3: (A) Co-occurrence keyword co-occurrence mapping. (B) Clustering keyword mapping.**

acceptance of telehealth [7] , providing valuable insights into the practical implications of telecare. As the field continues to evolve, future researchers are encouraged to build upon the foundations laid by these influential scholars, conducting more extensive and in-depth investigations. The current landscape reveals that research institutions in telecare are predominantly universities, indicating that the field is still in an exploratory stage with a nascent business model and operational system. To overcome these challenges and foster sustainable growth, prioritizing interdisciplinary and intersectoral cooperation is crucial, facilitating innovative advancements that align with evolving societal needs. The prevalent publication outlets for telecare research are high-quality journals within the healthcare science and services domain. This underscores the pivotal role of telecare in shaping the landscape of modern healthcare development. Looking ahead, researchers are urged to intensify their focus on journal literature within this field, with an emphasis on integrating telecare seamlessly into traditional healthcare models. This concerted effort will not only contribute to the scholarly discourse but also accelerate the practical integration of telecare into mainstream healthcare practices, ensuring its relevance and impact in the evolving healthcare landscape.

## 4.2 Analysis of research hotpots

*4.2.1 Telecare technology optimization.* Cluster #2 authentication and high-frequency keywords such as "telecare medicine information system" indicates that the development of user authentication schemes for telemedicine information systems is a prominent research topic in this field. Telecare information systems handle sensitive personal health information of patients, making it crucial to ensure healthcare quality and patient privacy through legitimate authorized access via user authentication. This helps prevent unauthorized users or malicious attackers from unauthorized access, data theft, or tampering [8]. Research in this area has highlighted the challenge of constructing user authentication schemes that balance security and privacy with the convenience and usability requirements of different users and adaptability to various scenarios and devices [9]. In response to this challenge, Radhakrishnan et al. proposed an efficient and secure dual authentication technique utilizing elliptic curve cryptography for smart cards. This technique achieves user anonymity, unforgeability, and session key negotiation [10]. Furthermore, future research should focus on advancing and innovating authentication technologies, including decentralized authentication, anomaly detection, and automated response. These areas present new possibilities for user authentication to effectively address evolving user information security needs.

Cluster #8, focused on artificial intelligence and related topics, indicates that the application of artificial intelligence in the field of telecare has garnered significant attention and recognition from researchers. The development of artificial intelligence technology has brought increased convenience and efficiency to telecare practices. By collecting substantial amounts of health data through intelligent monitoring devices and applications and utilizing machine learning and data analysis algorithms, artificial intelligence aids in identifying potential health problems, predicting disease trends, and providing timely health warnings and recommendations [11]. However, despite the potential benefits, the widespread adoption of AI technology in the telecare field still faces challenges. These include technological accessibility and the digital divide in less-developed regions, as well as a range of ethical and legal concerns. Therefore, future efforts should focus on strengthening AI infrastructure and establishing a robust ethical framework and legal regulations. This ensures that AI systems make fair and transparent decisions while upholding medical ethical principles and protecting privacy.

*4.2.2 Telecare application areas.* The identified clusters, specifically #1 emergency medicine, #4 asthma, and #7 emergency department, underscore the pivotal role of telecare in both emergency medicine and chronic disease management. In critical situations, such as emergencies, the swift and effective response becomes paramount for patient survival. Telecare emerges as a transformative solution, facilitating remote expert assistance through video consultations and continuous remote monitoring. This innovative approach not only ensures timely diagnosis but also enables remote experts to provide crucial treatment recommendations. By transcending geographical constraints inherent in traditional healthcare models, telecare optimizes decision-making processes and treatment plans for local emergency medical teams [12]. Chronic diseases, exemplified by cluster #4 asthma, present ongoing challenges for patients, necessitating continuous management and monitoring. Telecare emerges as a promising solution by reducing the frequency of hospital visits, thus saving time and lowering overall healthcare costs. An exemplary study by Mammen, JR et al. demonstrated the implementation of remote monitoring for asthma patients through smartphones. In addition to monitoring, patients were provided with self-management training. This holistic approach not only effectively controlled acute asthma exacerbations but also significantly enhanced the overall quality of life for the patients involve [13]. Furthermore, the emergence of cluster #3 COVID-19 suggests a notable acceleration in the development and utilization of telecare, especially in response to the global COVID-19 pandemic. Researchers such as Gilkey, MB et al. have illuminated the role of telecare as a viable alternative to traditional in-person care. By minimizing the need for face-to-face contact, telecare not only reduces the risk of contagion but also plays a crucial role in containing the spread of outbreaks [14]. In essence, these diverse clusters in telecare research highlight its adaptability and efficacy across healthcare domains, ranging from emergency medicine to chronic disease management and pandemic response. Telecare stands out as a versatile tool, revolutionizing healthcare delivery and asserting itself as a transformative force in the contemporary healthcare landscape. Future research should delve deeper into the nuanced applications of telecare across diverse medical contexts.

*4.2.3 Telecare goal evaluation.* Clusters #0 self-care, #5 telephone nursing, #9 distance learning, and the high-frequency keywords "management" and "impact" indicate that one of the goals of telecare is to enhance patients' self-care capabilities. Telecare facilitates patient education about diseases, improving their health literacy and self-management skills through online educational courses and interactive applications [15]. Additionally, patients can monitor their blood pressure, blood glucose, and other indicators by themselves. The collected monitoring data is automatically uploaded to the cloud for storage and analysis. Healthcare professionals can provide health guidance through telephone follow-ups and other methods. Self-monitoring and tracking empower patients to have a better understanding of their health status and take appropriate health promotion measures [16]. However, it is important to note that telecare may not be suitable for all types of diseases or patients. Some patients may face barriers to technology use or encounter language-related challenges that limit the benefits they can derive from telecare. In the future, community collaboration mechanisms could be considered, involving community workers and family members to ensure that patients receive optimal care and monitoring.

## 5 CONCLUSION

This study utilizes bibliometric methods to visualize and analyze the current research status and hot topics in the field of telecare. Telecare research has made significant progress over the past two decades, highlighting its global demand for addressing healthcare disparities. Focus areas include authentication solutions and artificial intelligence applications, with the aim of enhancing information security and improving healthcare outcomes. Telecare plays a crucial role in emergency medicine, chronic disease management, and self-care, and its development has been accelerated by the COVID-19 pandemic. Moving forward, it is important to prioritize interdisciplinary cooperation and consider community collaboration to deliver optimal healthcare.

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